CSCI 1851: Machine Learning for Biology and Health

Final Course Project

February 2026

1 Purpose

The final course project is intended to help you learn the course material through practice. It is designed to introduce the considerations involved in conducting an interdisciplinary research project through hands-on experience. The hope is that once you are familiar with what this work entails, you will be motivated to pursue research in this field. Through this project, you will apply machine learning models to the proposed multi-modal task of predicting diagnostic categories from images and clinical data. You will have the final presentation to share your work with your peers through presentations and a written report.

This type of project-based exercise will help you in developing skills like -

1. Thinking critically about the health task.
2. Exploring different methods to solve the task.
3. Designing robust experiments to validate your hypotheses.
4. Adapting your method to real-world datasets/challenges.
5. Reporting results in a clear and concise manner.

1.1 Problem Setting

In clinical settings, diagnostic decisions often rely on multiple sources of information. Visual inspection alone may not be sufficient; patient-specific metadata can also play an important role. In this project, you will develop a model that combines: Image features extracted from dermoscopic photographs and Structured clinical variables (age, sex, anatomical location). The goal is to predict one of seven diagnostic categories. The dataset is intentionally imbalanced to reflect real-world conditions. Robust modeling and validation strategies are expected.

2 Project details

2.1 Kaggle competition

With your team, you will work on a course project that will apply a multi-modal machine learning model to the task of predicting diagnostic categories from images and clinical data. This project is modeled as a Kaggle competition (link posted on EdStem). The description of the task and related datasets are provided on the Kaggle website.

Each team will develop and apply a machine learning framework to solve the prediction task using the released dataset. Around the mid-semester mark, the teams will submit their predictions, and we will release the leaderboard performance of the submitted predictions. For the finals, the teams will present their methods in presentations, submit their predictions/code, and write a report. We will test the models on a held-out dataset and publish a scoreboard ranking their final performances.

2.1.1 Interpretation Component for the FINAL portion only

For the final presentation and report, we would like you to include an analysis using any interpretation techniques we discussed in the course to interpret features the model is learning. Bonus points will be given for connecting your interpretation results to the existing literature.

2.2 Computing resources

You all have access to CS department computing resources as well as Brown University's computing resource, OSCAR.

2.3 Presentations and submissions

2.3.1 Model and code submission for Mid-term (20 points)

Each team should create a **private GitHub repository** for the project. This will allow us to track the contributions of the team members as well as access the code and the model for grading in the mid-term phase. Please ensure that you have a complete README in the GitHub repository so that we can download and run your models.
Grading Criteria: We will load and run your trained models on a selected set of held-out datasets, and also look at your code for completion and correctness of the implementation. Points will be deducted if we have trouble running your code/model or find any major logic flaw.

2.3.2 Final Model and code submission (20 points)

Each team should update the **private GitHub repository** created for the mid-term phase of the project. Please highlight the major changes made after

the mid-term project via README or comments in code. This will allow us to track the contributions of the team members as well as access the code and the model for grading in the final phase. Please ensure that you have a complete README in the GitHub repository so that we can download and run your models.
Grading Criteria: We will load and run your trained models on a selected set of held-out datasets, and also look at your code for completion and correctness of the implementation. Points will be deducted if we have trouble running your code/model or find any major logic flaw.

2.3.3 Final Project Presentation (30 points)

Each team will get a 7-minute slot (5 min for presentation + 2 min for questions) to present their method and results. We recommend the teams to practice their presentation timing beforehand to avoid going beyond the 5-min mark ($\sim$ 5 slides). In your presentation slides, please describe the following - model architecture, experimental design, and the results. The teams should be able to explain their choices for model development. They should include good figures/tables to summarize their results. Interesting observations and their explanations are highly encouraged.
Grading Criteria: The presentations will be graded on the content and how well you answer the questions.

2.3.4 Final Project Report (20 points)

The project report will include the following sections: Introduction, Method, Experiment Setup, and Results. The Introduction section should not exceed 200 words and summarize the overall work done by the team. The Method section will expand on the model architecture and the rationale behind different choices. The Experiment Setup will provide details on running and training the model. Finally, the observations will be reported in the Results section. The overall length of the report must not exceed 4 pages (single space, 12 point font), including write-up, figures, tables, and bibliography.

For the final, please include an analysis using one or more of the interpretation techniques we discuss in the course to interpret features the model is learning.
Grading Criteria: This submission will be reviewed by the instructor. Its evaluation will be based on the content (like description of various sections), structure, flow of logic, and the quality of writing and figures/tables.

2.4 Overall Participation (10 points)

A fraction of the score (10 points) will also be assigned for each student's contribution to the project as assessed by their team member. Note that the instructor can reduce the overall project score for a student if it is determined that they did not contribute substantially to the project.

2.5 Important dates

Date	**Task**	**Score**
Feb 24 (Tue)	Kaggle Competition is released	N/A
March 31 (Tue)	Submission of mid-term predictions/code ready to be evaluated	20
April 02 (Thu)	Leaderboard released and Test set 1 released	N/A
April 29 (Tue)	Submission of final predictions/code ready to be evaluated	20
April 30 (Thu)	Leaderboard released and Test set 2 released	N/A
May 05 (Tue)	Final project presentations	30
May 12 (Tue)	Reports due at 5 PM	20

2.6 Project policies

Collaboration Policy: Discussion of material with your classmates is both permitted and encouraged. However, showing, copying, or other sharing of answers to written questions and actual code with students outside your team on the project is not allowed. This includes publishing projects publicly on Github or any other public platform. In addition, reusing code or pre-trained models from another student outside your team or any public platform is not allowed.

Missed deadlines or late submissions: For all final project-related submissions, you can get a 48-hour extension for at most 2 deadlines without penalty. Excluding the scenario mentioned above, 20% of the total points will be deducted for late submissions and missed submissions won't be assigned any score. For project presentations, if you are unable to present on a particular day (due to travel, illness etc.), please inform the instructor at least 24 hours in advance. No-show on the day of your assigned presentation will be treated as a missed assignment.